



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

**UNIT 10 • STANDARD 6.G.2**

# Volume with Whole Number Edges

Lesson 10-1 • Teacher Copy

**Name:** Type your name      **Date:** Today's date      **Class:** Class period

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## Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.



unit cubes

*A box  $3 \times 2 \times 4$  holds 24 unit cubes, so  $V = 24$  cubic units*

### Volume

Write the definition:

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length  $\times$  width  $\times$  height

*A cereal box or shoe box — it has length, width, and height*

### Rectangular prism

Write the definition:

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one cubic unit

*A tiny cube that is  $1 \text{ in} \times 1 \text{ in} \times 1 \text{ in} = 1 \text{ in}^3$  (one cubic inch)*

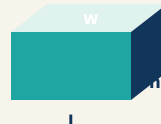
### Cubic units

Write the definition:

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*$V = l \times w \times h$ : for a box  $5 \times 3 \times 2$ , volume = 30 cubic units*

### Length, width, height

Write the definition:

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**flattened faces**

*A cross-shaped pattern of 6 rectangles folds into a rectangular box*

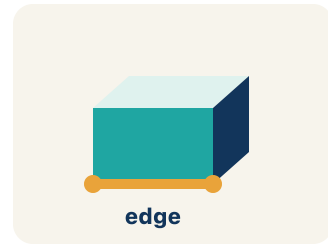
### **Net**

**Write the definition:**

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**edge**

*A cube has 12 edges — 4 along the top, 4 along the bottom, 4 vertical*

### **Edge**

**Write the definition:**

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## Guided Notes

Level 2 Standard



### WHAT WE'RE LEARNING TODAY

**I can find the volume of a rectangular prism with whole-number edges using  $\text{length} \times \text{width} \times \text{height}$ .**



**Fill in each blank as we go. Use the Word Bank to help you.**



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Volume

Rectangular prism

Cubic units

Length, width, height

Net

Edge



*Tap any word to see what it means and a picture.*

**1** The amount of space inside a three-dimensional solid is its

**2** A solid with six rectangular faces, like a box, is a .

**3** The unit used to measure volume, like cubic centimeters, is

**4** The three edge measurements of a rectangular prism are its

**5** A flat pattern that folds up into a solid is a .

**6** A line segment where two faces of a solid meet is an .



### Watch & Try — Worked Examples

*See the notes in action: watch one worked all the way through, then try the next with the same steps.*

 **I do — watch**

Follow each step as your teacher solves it.

**Problem:** What is the volume of a rectangular prism with  $l = 7$  in,  $w = 3$  in,  $h = 4$  in?

- A.  $84 \text{ in}^3$
- B.  $14 \text{ in}^3$
- C.  $84 \text{ in}^2$
- D.  $42 \text{ in}^3$

**Step 1**  $V = l \times w \times h = 7 \times 3 \times 4 = 84$  cubic inches.

 **Answer:** A.  $84 \text{ in}^3$

 **We do — together**

Solve this one with your class using the same steps.

**Problem:** A time capsule box has a volume of  $120 \text{ cm}^3$ . Its length is 10 cm and width is 4 cm. What is its height?

- A. 3 cm
- B. 6 cm
- C. 4 cm
- D. 12 cm

**Step 1** \_\_\_\_\_

**Step 2** \_\_\_\_\_

**Answer:** \_\_\_\_\_

**You do — your turn**

Now try one on your own. Show every step.

**Problem:** Which unit is used for volume?

- A. Cubic inches ( $\text{in}^3$ )
- B. Square inches ( $\text{in}^2$ )
- C. Inches (in)
- D. Degrees ( $^\circ$ )

Show your work:

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## Try It

Solve on your own. Check the answer key when you are done.

**1. A time capsule box is 5 units long, 3 units wide, and 2 units tall. How many unit cubes fill it?**

- A. 30 cubic units
- B. 10 cubic units
- C. 15 cubic units
- D. 25 cubic units

Show your work:

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**2. You stack unit cubes in a capsule: each layer has 6 cubes, and there are 4 layers. What is the volume?**

- A. 24 cubic units
- B. 10 cubic units
- C. 12 cubic units
- D. 18 cubic units

Show your work:

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## Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

**A box needs to hold exactly 60 cubic inches. Give three different sets of whole-number dimensions that work. Which set would make the box closest to a cube shape? Why might that matter?**

*Sentence starter: Option 1:  $\underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$ . Option 2:  $\underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$ . Option 3:  $\underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$ .*

*The  $\underline{\hspace{1cm}}$  option is closest to a cube because  $\underline{\hspace{1cm}}$ . This matters because  $\underline{\hspace{1cm}}$ .*

Show your work:

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## Reflect — Exit Ticket

**A rectangular prism has  $l = 11$  in,  $w = 5$  in,  $h = 4$  in. What is the volume?**

- A.  $220 \text{ in}^3$
- B.  $55 \text{ in}^3$
- C.  $220 \text{ in}^2$
- D.  $200 \text{ in}^3$

Your answer:

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## Answer Key & Teacher Guide

1. **Notes 1:** Volume
2. **Notes 2:** Rectangular prism
3. **Notes 3:** Cubic units
4. **Notes 4:** Length, width, height
5. **Notes 5:** Net
6. **Notes 6:** Edge
7. **We Try (notes):** A. 3 cm —  $V = l \times w \times h \rightarrow 120 = 10 \times 4 \times h \rightarrow 120 = 40h \rightarrow h = 3 \text{ cm}$ .
8. **You Try (notes):** A. Cubic inches ( $\text{in}^3$ ) — *Volume measures 3D space, so it uses cubic units like  $\text{in}^3$ . Square units ( $\text{in}^2$ ) are for area, and inches are for length.*
9. **Try It 1:** A. 30 cubic units —  $V = \text{length} \times \text{width} \times \text{height} = 5 \times 3 \times 2 = 30 \text{ cubic units}$ .
10. **Try It 2:** A. 24 cubic units — *Each layer has 6 cubes and there are 4 layers, so  $6 \times 4 = 24$  cubic units. This is the same as  $\text{length} \times \text{width} \times \text{height}$  with one factor being the number of layers.*
11. **Exit Ticket:** A.  $220 \text{ in}^3$  —  $V = 11 \times 5 \times 4 = 220 \text{ cubic inches}$ . Remember: volume uses cubic units ( $\text{in}^3$ ).